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Cambridge IGCSE Combined and Co-ordinated Sciences

B3 Biological Molecules

Comprehensive Exam Preparation Guide

B3 Biological Molecules — Cambridge IGCSE Biology Study Guide

Chapter Overview

In this chapter you will:

- Find out about the components of carbohydrates, proteins and lipids
- Use food tests to detect the presence of different biological molecules

All learners study all content in this chapter. The whole chapter is one topic: B3.01 Carbohydrates, fats and proteins.

Important Terms

Term	Definition
Biological molecules	Molecules that make up living cells, such as carbohydrates, fats and proteins; they are not often found in non-living things
Element	A substance made of only one kind of atom, e.g. carbon (C), hydrogen (H), oxygen (O) or nitrogen (N)
Compound	A substance made of two or more different elements chemically combined, e.g. glucose
Molecule	A particle made up of two or more atoms combined together, e.g. O_2 or $C_6H_{12}O_6$
Carbohydrates	Substances that include sugars, starch and cellulose; they contain carbon, hydrogen and oxygen, with about twice as many hydrogen atoms as carbon or oxygen atoms
Sugars	Carbohydrates that have relatively small molecules; they are soluble in water and they taste sweet
Glucose	A sugar that is used in respiration to release energy; formula $C_6H_{12}O_6$; the form in which carbohydrate is transported in the blood
Reducing sugar	A sugar, such as glucose, which turns Benedict's solution orange-red when they are heated together
Starch	A carbohydrate that is used as an energy store in plant cells; made of many glucose molecules in a long chain that coils into a spiral
Glycogen	A carbohydrate that is used as an energy store in animal cells; made of many glucose molecules in branched chains
Cellulose	A carbohydrate found in plants, where it makes up the cell walls; it is also made from glucose molecules
Lipids	Substances containing carbon, hydrogen and oxygen; they are insoluble in water and are used as energy stores in organisms; another name for fats and oils
Fats	Lipids that are solid at room temperature
Oils	Lipids that are liquid at room temperature

Glycerol	One of the two kinds of smaller molecule that make up a fat; each fat molecule contains one glycerol molecule
Fatty acids	Smaller molecules that combine with glycerol to make fats and oils; each fat molecule contains three fatty acid molecules
Protein	A substance whose molecules are made of many amino acids linked together; each different protein has a different sequence of amino acids
Amino acids	Substances with molecules containing carbon, hydrogen, oxygen and nitrogen; there are 20 different amino acids found in organisms
Enzymes	Proteins that act as biological catalysts, speeding up chemical reactions in living organisms; all enzymes are proteins
Antibodies	Molecules secreted by white blood cells, which bind to pathogens and help to destroy them; antibodies are proteins
Pathogens	Microorganisms, such as bacteria, that cause disease
Haemoglobin	The red pigment that transports oxygen in mammalian blood; it is a protein
Keratin	The protein that forms hair (and fingernails)
Iodine solution	A solution of iodine in potassium iodide; it is orange-brown, and turns blue-black when mixed with starch
Benedict's solution	A blue liquid that turns orange-red when heated with a reducing sugar
Biuret reagent	A blue solution (also called biuret solution) that turns violet (purple) when mixed with proteins
Emulsion	A liquid containing two substances that do not fully mix; one of them forms tiny droplets dispersed throughout the other
Iodine test	The food test for starch: add iodine solution to the food; a blue-black colour means starch is present
Benedict's test	The food test for reducing sugars: heat the food with Benedict's solution in a water-bath; green, yellow or orange-red means reducing sugar is present
Biuret test	The food test for proteins: add biuret solution to the food and shake, with no heating; a violet (purple) colour means protein is present
Ethanol emulsion test	The food test for fats and oils: shake the food with ethanol, then pour the ethanol into water; a milky white emulsion means fat or oil is present
Water-bath	A container of water kept at a steady temperature, used to heat test-tubes gently and evenly, e.g. at about 80 °C for the Benedict's test

B3.01 Carbohydrates, Fats and Proteins

Before You Start: Elements, Compounds, Atoms and Molecules

The chapter begins by checking some chemistry ideas you need for this topic. Biological molecules are compounds made from just a few elements, so make sure you are clear about these words:

- **Atom** — the smallest particle of an element

- **Element** — a substance made of only one kind of atom, such as carbon (C), hydrogen (H), oxygen (O) or nitrogen (N)
- **Molecule** — a particle made up of two or more atoms combined together
- **Compound** — a substance made of two or more different elements chemically combined

Answers to the Before You Start questions:

Before You Start question	Answer
1a Oxygen gas is made up of pairs of oxygen atoms combined together. What is the formula for oxygen gas?	O_2
1b What do we call a particle made up of two or more atoms combined together?	A molecule
1c Is oxygen an element or a compound?	An element, because it contains only one kind of atom (oxygen atoms)
2a Glucose is made up of the elements that have the symbols C, H and O. What do C, H and O stand for?	Carbon, hydrogen and oxygen
2b One molecule of glucose contains 6 atoms of carbon, 12 atoms of hydrogen and 6 atoms of oxygen. Can you write the formula for glucose?	$C_6H_{12}O_6$
2c Is glucose an element, a compound or a mixture?	A compound, because three different elements are chemically joined together, in the same fixed proportions in every molecule

Science in Context: How Did Life on Earth Start?

One of the greatest mysteries of life on Earth is how it began. Living cells are made up of biological molecules, and these are not often found in non-living things. Scientists think that these molecules may have formed from simpler substances when the Earth was still very young, perhaps 4 billion years ago. The mixture of chemicals in the seas and atmosphere at that time, together with the energy provided by volcanic eruptions and lightning strikes, could possibly explain how the first molecules of life were produced.

Another possibility is that at least some of these molecules arrived on Earth from outer space. Meteorites are rocks that arrive on Earth from space, and they may have delivered molecules such as amino acids (the building blocks of proteins). If we can understand how life first arose from biological molecules, this could help us to answer another important question: is there life anywhere else in the universe?

Q: Discussion question 1: Most scientists think that all life on Earth today comes from other life. Suggest why scientists think it is unlikely that life could arise from non-living things today.

A: Conditions on Earth today are very different from those about 4 billion years ago. The atmosphere now contains a lot of oxygen, which reacts with and breaks down simple carbon compounds, and the mixture of chemicals in the seas and air has changed. The Earth is also full of living organisms, especially microorganisms such as bacteria, which would quickly use any new biological molecules as food or break them down before they could build up and join together. Finally, getting from simple molecules to even the simplest living cell probably took millions of years, so we would not expect to see it happen now.

What Is a Human Made Of?

The most abundant compound in our bodies is water. Almost all of the rest is made of the three main types of organic compound that make up the bodies of all living things: **carbohydrates, fats and proteins**.

Substance	Approximate percentage of total body mass
Water	62%
Protein	16%
Fat	16%
Carbohydrate	1%
Calcium and phosphorus (mostly in bones), plus other elements such as potassium, sodium, phosphate and chloride ions	5%

Notice how little of the body is carbohydrate. We use carbohydrate (glucose) quickly in respiration and store only a small amount of it, while fat forms a much bigger energy store.

Carbohydrates

- **Carbohydrates** include starch, cellulose and sugars
- Carbohydrate molecules are made from three elements: **carbon, hydrogen and oxygen** (C, H and O)
- One molecule of carbohydrate contains **about twice as many hydrogen atoms as carbon or oxygen atoms**. For example, glucose ($C_6H_{12}O_6$) has 12 hydrogen atoms but only 6 carbon atoms and 6 oxygen atoms

Sugars and Glucose

- **Sugars** are the simplest types of carbohydrate, with the smallest molecules. Sugars taste sweet, and they are soluble in water
- **Glucose** is one type of sugar. A glucose molecule is made of 6 carbon atoms, 12 hydrogen atoms and 6 oxygen atoms, so its formula is $C_6H_{12}O_6$
- **Shape**: the atoms in a glucose molecule are arranged to form a hexagon shape. Figure B3.01 shows glucose molecules drawn as small hexagons, because glucose molecules are quite small
- **Transport**: glucose is the form in which carbohydrates are transported around the human body. Glucose dissolves in blood plasma, which delivers it to every cell
- **Energy**: cells need glucose for energy, because glucose is used in respiration to release energy

Starch and Glycogen: Big Molecules Made from Glucose

Glucose molecules can link together in chains to form much larger molecules. Animals and plants do this in slightly different ways:

- **Glycogen** (in animals): glucose molecules link together to form glycogen, which is used as an energy store in animal cells. The liver helps to keep the concentration of glucose in the blood constant (you can read about this in Chapter B9)
- **Starch** (in plants): glucose molecules are linked together in a slightly different way to make starch, which is stored in plant cells. Starch can be broken down to form glucose again when the plant needs it
- **Cellulose**, which makes up plant cell walls, is another plant carbohydrate made from glucose molecules

Figure B3.02 shows a small part of a starch molecule. It is made from many glucose molecules (drawn as hexagons) linked together in a long chain, which coils into a spiral shape. Glycogen is also made of chains of glucose molecules, but its chains are branched rather than coiled.

The key idea to learn:

- many glucose molecules → starch (the energy store in plants)
- many glucose molecules → glycogen (the energy store in animals)

Carbohydrate	Size of molecule	Made from	Found in	Job
Glucose	Small (a sugar)	It is a single unit, the building block for the others	Plants and animals	Transported in blood plasma to cells; used in respiration to release energy
Starch	Very large	Many glucose molecules in a long chain that coils into a spiral	Plants, not animals	Energy store in plant cells
Glycogen	Very large	Many glucose molecules in branched chains	Animals, not plants	Energy store in animal cells, e.g. in the liver
Cellulose	Very large	Many glucose molecules	Plants, not animals	Makes up plant cell walls

Testing for Starch: The Iodine Test

- **Iodine solution** is a solution of iodine in potassium iodide. It is orange-brown
- **Positive result**: iodine solution turns blue-black when it is mixed with starch
- **Negative result**: it stays orange-brown if there is no starch
- **No heating** is needed: you just add a drop or two of iodine solution to a piece of food on a white tile

Figure B3.03 shows iodine solution being dropped onto a cut potato. The blue-black colour shows that starch is present in the potato.

Iodine test: orange-brown → blue-black means starch is present.

Testing for Reducing Sugars: Benedict's Test

- **Benedict's solution** is blue. It changes colour when it is heated with reducing sugars, such as glucose
- **Most sugars** can be detected using Benedict's solution
- **Heat is essential**: the mixture must be heated in a water-bath (about 80 °C), otherwise there is no colour change
- **Dissolve first**: the food is cut into small pieces and shaken with a little water, so that any sugar dissolves

The colour you get depends on the **concentration** of reducing sugar:

Colour after heating with Benedict's solution	Amount of reducing sugar
Blue (stays the same colour)	None, so the result is negative
Green	Very low concentration
Yellow	Slightly higher concentration
Orange-red (often called brick red)	Very high concentration

However, even low concentrations of glucose will eventually turn the blue colour to orange-red if they are heated for long enough. So, to compare foods fairly, heat them all for the same length of time, or time how long each one takes to change colour.

Figure B3.04 shows positive results of the Benedict's test. The tube on the left contains a small amount of reducing sugar, and the blue Benedict's solution has changed to green. The tube on the right

contains a large amount of reducing sugar, and the mixture has turned orange-red.

Benedict's test: blue → green → yellow → orange-red as the amount of reducing sugar increases.

Good to know: not every sugar is a reducing sugar. Ordinary table sugar (sucrose) is not a reducing sugar, so it leaves Benedict's solution blue. This is why the book says that most sugars, rather than all sugars, can be detected with Benedict's solution.

Using a Water-Bath

The Benedict's test needs heat, and the safest way to heat the tubes is to stand them in a water-bath:

- **Electric water-bath**: can be set to a temperature above room temperature (about 80 °C for the Benedict's test)
- **Home-made water-bath**: heat water in a beaker over a burner, and stand the test-tube in the hot water (Figure B3.05)
- **0 °C water-bath**: fill a beaker with crushed ice
- **Keeping the temperature steady**: check the temperature with a thermometer, and either heat with the burner or add cold water to keep the temperature fairly constant

Figure B3.05 shows a boiling tube containing a mixture of Benedict's solution and food, standing in a beaker of water that is being heated over a burner. A water-bath heats the mixture gently and evenly; heating a tube directly in a flame could make the liquid boil suddenly and spit out.

Experimental Skills B3.01: Testing for Carbohydrates

Aim

To test several different foods to find out if they contain starch or reducing sugar.

You Will Need

- A selection of foods to test, kept separate from one another
- A knife to cut the foods into small pieces
- A white tile
- A small bottle of iodine solution, with a dropper
- A water-bath set at about 80 °C (an electric one, or one you make by heating water in a beaker over a burner, as in Figure B3.05)
- A thermometer to check the temperature of the water-bath
- Several large test-tubes (boiling tubes), and a beaker to stand them in
- Some Benedict's solution, with a dropper

Safety

- Take care when using a knife to cut food (this is the safety point printed in the book)
- Also sensible: wear eye protection, take care not to burn or scald yourself on the burner or the hot water, and point boiling tubes away from people

Method: Testing for Starch

1. Read through the instructions, and decide which foods you are going to test for starch.
2. Construct a results chart ready to fill in as you work, with a column for the food, a column for your observations and a column for your conclusion. Write down the colour that you see; do not write "no change".
3. Take a small piece of the first food and place it on the white tile. Add a drop or two of iodine solution. If the iodine solution stays brown, there is no starch in the food; if you see a dark blue or black colour, then starch is present.
4. Repeat with several other samples of different foods.

Method: Testing for Reducing Sugars

1. Read through the instructions and construct a results chart.
2. Take some of your first food and cut it into very small pieces. Put the pieces into a boiling tube and add a little water. Shake the tube to try to dissolve any sugar in the food.
3. Add enough Benedict's solution to cover the food. Shake the tube to mix the contents.
4. If you are using an electric water-bath, fill a beaker with water from the water-bath, stand the beaker in the water-bath and stand the test-tube in the beaker. If you have made your own water-bath, simply stand the test-tube in the beaker of hot water (Figure B3.05).
5. Watch as the mixture of food and Benedict's solution heats up. If it stays blue, there is no reducing sugar in the food; if it changes to red, there is reducing sugar in the food.
6. Repeat with several other samples of different foods.

Why these steps matter: cutting the food into very small pieces helps the substances inside it to dissolve in the water or reach the reagent, and the white tile makes the colour easy to see. Writing "no change" is not acceptable, because it does not tell anyone what colour you actually saw.

What the Results Look Like

Test	Negative result (substance absent)	Positive result (substance present)
Iodine test for starch	Iodine solution stays orange-brown	Dark blue or black (blue-black)
Benedict's test for reducing sugar	Stays blue after heating	Green, yellow or orange-red after heating, depending on how much reducing sugar there is

An example results chart (your own results depend on the foods you test):

Food	Test	Observation (colour seen)	Conclusion
Potato	Iodine	Blue-black	Starch present
Grapes	Iodine	Orange-brown	No starch
Grapes	Benedict's	Orange-red after heating	Large amount of reducing sugar
Cooking oil	Benedict's	Blue after heating	No reducing sugar

Conclusion

Foods that turn iodine solution blue-black contain starch. Foods that turn Benedict's solution green, yellow or orange-red when heated contain reducing sugar; the further the colour moves along the series towards orange-red, the more reducing sugar the food contains.

Answers to the Questions in This Box

Q: 1 Make a list of the foods that contained starch, and the foods that contained reducing sugar.

A: The list depends on which foods you tested. Typical results are: starch in potato, bread, rice, pasta and flour; reducing sugar in grapes, apples, onions, honey, fruit juice and milk. Foods such as cooking oil and table sugar (sucrose) give a negative result in both tests.

Q: 2 Do these foods come from plants, or from animals?

A: Most of them come from plants. Starch is the energy store in plant cells, so starchy foods such as potatoes, rice and wheat flour (used to make bread and pasta) all come from plants. Animals store carbohydrate as glycogen instead, so foods that come straight from animals, such as meat and eggs, do not contain starch. Most foods with reducing sugar also come from plants, such as fruits and onions (honey is made by bees from the sugary nectar of flowers). Milk is an exception: it comes from animals and contains a reducing sugar called lactose.

Activity B3.01: Comparing Reducing Sugar Content

When you tested foods for reducing sugar, you probably noticed that the red colour took a few minutes to appear. **The less time it takes for the red colour to appear, the more reducing sugar there is in the food.** In this activity (in a group of two or three), you discuss how to use this fact to compare the quantity of reducing sugar in three different foods, and write a list of instructions that someone else could follow.

A model set of instructions:

1. Choose three foods, for example grapes, apple and onion. Weigh out the same mass of each food (for example 2 g) and cut each one into equally small pieces.
2. Put each food into its own labelled boiling tube, add the same volume of water (for example 5 cm^3) to each, and shake each tube for the same length of time.
3. Add the same volume of Benedict's solution (for example 2 cm^3) to each tube and shake to mix.
4. Stand all three tubes in the same water-bath at about 80 °C at the same moment, and start a stopwatch.
5. Record the time taken for each mixture to turn red. If a tube has not turned red after 10 minutes, record "more than 10 minutes".
6. Repeat the whole experiment twice more and calculate a mean time for each food.
7. The food with the shortest mean time contains the most reducing sugar, and the food with the longest time contains the least.

Variables to mention in your instructions:

- **Independent variable** (what you change): the type of food
- **Dependent variable** (what you measure): the time taken for the red colour to appear
- **Controlled variables** (what you keep the same): mass of food, size of the pieces, volume of water, volume of Benedict's solution and the temperature of the water-bath

Peer assessment: swap your instructions with another group. Check whether their instructions are easy to follow and whether they would correctly show which food contains the most reducing sugar. Give feedback on what the group did well and one or two things that could be improved. Then think carefully about the feedback you receive, and what you would do differently if you planned a similar experiment in future.

Fats and Oils (Lipids)

Fats and oils are also known as **lipids**:

- A **fat** is a lipid that is **solid** at room temperature (for example, butter)
- An **oil** is a lipid that is **liquid** at room temperature (for example, olive oil)
- Lipids are **insoluble in water**

Fats and oils contain the same three elements as carbohydrates: **carbon, hydrogen and oxygen**. However, in fats each molecule contains **much less oxygen** than is found in a carbohydrate molecule.

Fat molecules are made of two kinds of smaller molecule: **glycerol** and **fatty acids**. Figure B3.06 shows a glycerol molecule drawn as a short upright bar and a fatty acid molecule drawn as a long zigzag line. In the fat molecule, **three fatty acid molecules are joined to one glycerol molecule**.

1 glycerol + 3 fatty acids → 1 fat (lipid) molecule

Why Organisms Need Fats

- **Cell membranes**: fats are important for making cell membranes
- **Energy stores**: fats are used as energy stores, and fat stores contain more energy per gram than carbohydrate stores
- **Insulation**: mammals often have a layer of cells containing fat droplets beneath the skin. This fat-containing tissue is both an energy store and a heat insulating layer

Figure B3.08 shows a bearded seal. Bearded seals have a really thick layer of fat underneath their skin, which helps them to survive in icy Arctic waters.

Testing for Fats and Oils: The Ethanol Emulsion Test

Fats and oils do not dissolve in water, but they do dissolve in **ethanol**. We can use this to detect them in food:

1. The food is cut into small pieces and shaken with ethanol. Any fats in the food dissolve in the ethanol.
2. The ethanol is poured into a clean tube containing water.
3. If there are fats in the ethanol, they form tiny droplets in the water, which give it a milky appearance (Figure B3.07).

The mixture of tiny droplets in water is called an **emulsion**, so this test is called the **ethanol emulsion test**. The milky colour appears because the fat cannot stay dissolved once the ethanol mixes with the water, so it comes out as millions of tiny droplets that scatter light.

Figure B3.07 shows a test-tube with a cloudy, milky white layer: this is what a positive result looks like. If there is no fat in the food, the liquid stays clear.

Ethanol emulsion test: clear → milky white emulsion means fat or oil is present.

Experimental Skills B3.02: Testing for Fats and Oils

Aim

To test a selection of different foods to find out if they contain fats or oils.

You Will Need

- A selection of foods to test, kept separate from one another
- A knife to cut the foods into small pieces
- Several very clean test-tubes
- A small bottle of ethanol, with a dropper
- Some distilled water

Safety

- This box does not print a safety note, but ethanol is highly flammable: keep it well away from flames, so no Bunsen burners should be lit nearby
- Take care when using a knife to cut food, and wear eye protection

Method

1. Read through the instructions and construct a results chart.
2. Take some of your first food and cut it into very small pieces. Put the pieces into a very clean test-tube and add a little ethanol. Shake the tube to try to dissolve any fats or oils in the food.
3. Take another clean test-tube and put some distilled water into it. Very carefully, pour the ethanol from the first tube into the distilled water. Do not let any of the food go in!

If you see a milky white colour appearing, this shows that there was fat in the food. The tubes must be very clean, and no food must go into the water, because traces of grease or small bits of food could make the water look cloudy and give a false positive result.

What the Results Look Like

Observation	Conclusion
A milky white (cloudy white) emulsion forms	Positive: fat or oil is present
The liquid stays clear and colourless	Negative: no fat or oil

Conclusion

Foods that produce a milky white emulsion contain fats or oils; foods that leave the water clear do not contain any fat that the test can detect.

Answers to the Questions in This Box

Q: 1 Which food(s) gave a positive result? Which gave a negative result?

A: This depends on the foods you tested. Typically, butter, margarine, cooking oil, cheese, peanuts and crisps give a positive result (a milky white emulsion), while foods such as apple, table sugar, egg white and boiled potato give a negative result (the liquid stays clear).

Q: 2 Use your observations to suggest how you could estimate which of the types of food contains the most fats and/or oils?

A: Compare how milky (cloudy) the emulsion is in each tube: the more fat or oil the food contains, the thicker and whiter the emulsion. To make the comparison fair, use the same mass of each food, cut into pieces of the same size, with the same volume of ethanol and the same volume of water, shaken for the same time. You could then rank the tubes in order of cloudiness, for example by holding each tube in front of a card with black writing on it and seeing which tube hides the writing most.

Proteins

Protein molecules contain four elements: **carbon, hydrogen, oxygen and nitrogen** (C, H, O and N). Some proteins, but not all, also contain a small quantity of the element **sulfur**.

- **Amino acids**: a protein molecule is made of a long chain of smaller molecules called amino acids (Figure B3.09)
- **20 kinds**: there are 20 different amino acids
- **Precise order**: each kind of protein has different amino acids linked together in a precise order (sequence)
- **One change matters**: if even one amino acid is changed, or moves into a different place in the protein molecule, then we get a different protein
- **Endless variety**: so there is an almost infinite number of different kinds of protein that can be made

Think of the 20 amino acids like the letters of an alphabet. By arranging letters in different orders you can make a huge number of different words; by arranging amino acids in different orders, cells make a huge number of different proteins.

Figure B3.09 shows a small part of a protein molecule as a chain of coloured blocks. Each block is one amino acid molecule, and the different colours represent different kinds of amino acid.

Many amino acids (20 kinds) joined in a specific sequence → one protein molecule

Functions of Proteins

Proteins have many different functions in organisms:

Protein	Function
Enzymes	All enzymes are proteins; they are biological catalysts that speed up reactions in cells (you will study enzymes in the next chapter)
Antibodies	Help to protect the body against pathogens; they are secreted by white blood cells, bind to pathogens and help to destroy them
Haemoglobin	The red pigment that transports oxygen in mammalian blood
Proteins in cell membranes	Proteins are important for forming cell membranes in all organisms
Keratin	The protein that makes up hair and fingernails in humans

Testing for Proteins: The Biuret Test

- **Biuret reagent** (also called biuret solution) is blue
- **Positive result**: it changes to violet (purple) when it is in contact with proteins; a pale lilac colour means only a little protein
- **Negative result**: it stays blue if there is no protein
- **No heating** is needed: just add the biuret solution to the food and shake

Figure B3.10 shows negative and positive results of the biuret test. The tube on the left has no protein, so it is still blue; the tube on the right contains protein, so it has turned violet.

Biuret test: blue → violet (purple) means protein is present.

Good to know: biuret reagent is a mixture of dilute copper sulfate solution and sodium hydroxide solution, and some exam questions describe adding these two solutions separately. Whichever way it is described, in an exam always say that the biuret test is the test for protein.

Experimental Skills B3.03: Testing for Proteins

Aim

To test a selection of different foods to find out if they contain proteins.

Using Control Tubes

You may find it helpful to start by testing something that you know contains protein, such as cooked egg white. Keep a tube showing a **negative result** (biuret reagent added to water, which will be blue) next to the tube showing a **positive result** (violet) in a test-tube rack. You can then compare each of your food tests with these two tubes to decide whether the result is negative or positive.

You Will Need

- A selection of foods to test, kept separate from one another
- A knife to cut the foods into small pieces
- Several clean test-tubes
- A small bottle of biuret solution with a dropper

Safety

- Take care when using a knife to cut food (this is the safety point printed in the book)
- Also sensible: biuret solution contains sodium hydroxide, which can harm skin and eyes, so wear eye protection and wash off any splashes with plenty of water

Method

1. Read through the instructions and construct a results chart.
2. Take some of your first food and cut it into very small pieces. Put the pieces into a clean test-tube and add some biuret solution. Shake the test-tube to mix the contents.
3. Look for a colour change from blue to violet.

What the Results Look Like

Observation	Conclusion
Stays blue, the same colour as the water control tube	Negative: no protein
Turns violet (purple), like the egg white tube	Positive: protein is present

Conclusion

Foods that turn biuret solution from blue to violet contain protein. Typical positive results come from egg white, meat, milk, cheese and beans.

Answers to the Questions in This Box

Q: 1 Do any of the foods you investigated contain no protein? How sure are you of this?

A: Any food that stayed blue, the same colour as the negative control tube, appeared to contain no protein; typical examples are cooking oil, table sugar and apple. However, we cannot be completely sure. The biuret test only shows protein if there is enough to give a visible colour change, so a food might contain a very small amount of protein that was not detected. A very pale violet can also be hard to tell apart from blue, especially if the food itself is coloured. We could be more confident by repeating the test, using more of the food, and comparing each tube carefully with the positive and negative control tubes against a white background.

Q: 2 Suggest one way you could improve the method to make the comparisons of the food types more valid.

A: Use the same mass of each food (for example, weigh 1 g of each), cut into pieces of the same size, and add the same volume of biuret solution to each tube, measured with a syringe or measuring cylinder rather than counted in drops. Then any difference in colour is caused only by the type of food. Other good improvements are to leave every tube for the same time before recording the colour, and to compare every tube with the same positive and negative control tubes.

Reflection: Remembering the Reagents

The book asks you to discuss with a partner good ways of remembering which reagent is used for which food test, and to decide which methods work best for you. Some ideas:

- Use a mnemonic: **"I Saw Big Sharks, Brave Penguins and Enormous Fish"** stands for Iodine for Starch, Benedict's for (reducing) Sugar, Biuret for Protein and Ethanol for Fat
- Remember **"Benedict's needs a bath"**: Benedict's is the only food test that is heated in a water-bath
- Remember **"proteins go purple"**: the biuret test turns purple (violet) when protein is present
- Picture the photos: the blue-black potato (starch), the orange-red tube (reducing sugar), the violet tube (protein) and the milky tube (fat)
- Make flashcards with the reagent on one side and the substance, method and colour change on the other, and test yourself regularly

Project B3.01: Modelling Biological Molecules

Working in a group of three or four, you make models of some biological molecules using paper and card. Different groups could make models of:

- **Starch and glycogen molecules**: both are made of glucose molecules linked in chains, but starch is coiled while glycogen is branched. Identical paper hexagons make good glucose molecules
- **Protein molecules**: made up of different sequences of amino acids. Use different colours or shapes for different amino acids, and show that changing the order makes a different protein
- **Fats and oils**: made of glycerol and fatty acids. Use one card strip for the glycerol, with three zigzag strips for the fatty acids attached to it

Comparing Carbohydrates, Fats and Proteins

Feature	Carbohydrates	Fats and oils (lipids)	Proteins
Elements	C, H, O (about twice as many H atoms as C or O atoms)	C, H, O (much less O than carbohydrates)	C, H, O, N (some also contain sulfur)
Smaller molecules they are made from	Starch and glycogen are made from glucose	Glycerol and fatty acids (1 glycerol + 3 fatty acids)	Amino acids (20 different kinds)

Examples	Glucose, starch, glycogen, cellulose	Butter (a fat), olive oil (an oil), fat under the skin	Enzymes, antibodies, haemoglobin, keratin
Uses in organisms	Glucose for respiration; starch (plants) and glycogen (animals) as energy stores	Cell membranes; energy stores; heat insulation	Enzymes, antibodies, oxygen transport, cell membranes, hair and nails
Food test	Iodine test (starch); Benedict's test (reducing sugars)	Ethanol emulsion test	Biuret test

Food Tests at a Glance

Food test	Tests for	What you do	Heat?	Negative result	Positive result
Iodine test	Starch	Add a drop or two of iodine solution to the food on a white tile	No	Stays orange-brown	Blue-black
Benedict's test	Reducing sugars, e.g. glucose	Shake the food with a little water, add Benedict's solution, then heat in a water-bath at about 80 °C	Yes	Stays blue	Green, yellow or orange-red, depending on the amount of reducing sugar
Biuret test	Proteins	Add biuret solution to the food and shake	No	Stays blue	Violet (purple)
Ethanol emulsion test	Fats and oils	Shake the food with ethanol, then pour the ethanol into water	No	Stays clear	Milky white emulsion

Key Formulas and Concepts

1. Glucose: $C_6H_{12}O_6$ — 6 carbon, 12 hydrogen and 6 oxygen atoms, arranged in a hexagon shape
2. Carbohydrates contain C, H and O, with about twice as many hydrogen atoms as carbon or oxygen atoms
3. Fats and oils contain C, H and O, but much less oxygen than carbohydrates
4. Proteins contain C, H, O and N (some also contain sulfur)
5. Many glucose molecules → starch (energy store in plants) or glycogen (energy store in animals)
6. 1 glycerol + 3 fatty acids → 1 fat or oil (lipid) molecule
7. Many amino acids (20 kinds) in a specific sequence → a protein; a different sequence gives a different protein
8. Iodine test for starch (no heating): orange-brown → blue-black
9. Benedict's test for reducing sugars (heat in a water-bath at about 80 °C): blue → green → yellow → orange-red as the concentration of reducing sugar increases
10. Biuret test for proteins (no heating): blue → violet (purple)
11. Ethanol emulsion test for fats and oils: shake the food with ethanol, pour the ethanol into water → milky white emulsion

12. Human body composition by mass: about 62% water, 16% protein, 16% fat, 1% carbohydrate and 5% other substances
13. Fat stores contain more energy per gram than carbohydrate stores (roughly twice as much)

Exam Focus Priorities

1. **Know the elements in each molecule** — carbohydrates and fats contain only C, H and O; proteins contain C, H, O and N. Nitrogen is the element found in all proteins but not in carbohydrates or fats.
2. **Link each large molecule to its smaller building blocks** — starch and glycogen are made from glucose; fats and oils from fatty acids and glycerol; proteins from amino acids. A common mistake is to say that fats are made of amino acids, or that starch is made of "sugar" without naming glucose.
3. **Starch is found in plants, glycogen in animals** — glucose is found in both. Starch is the energy store in plant cells and glycogen is the energy store in animal cells.
4. **Only the Benedict's test needs heating** — always write "heat in a water-bath (about 80 °C)". Never heat the biuret test; this is the error in Practice Question 4a(i).
5. **Always give the colour before and after** — iodine orange-brown → blue-black; Benedict's blue → orange-red; biuret blue → violet; ethanol test → milky white emulsion. Write the colour you actually see (e.g. "stayed blue"), never just "no change" or "positive".
6. **Do not mix up the colours** — iodine solution starts orange-brown, not blue. Benedict's and biuret both start blue, but Benedict's ends orange-red and biuret ends violet. Swapping the start and end colours, as in Practice Question 4a(ii), loses the mark.
7. **Describe the ethanol emulsion test in the right order** — shake the food with ethanol first, then pour the ethanol (not the food) into water; a milky white emulsion means fat. Use very clean tubes.
8. **Read results tables carefully** — "contains only fat" means positive for the ethanol emulsion test but negative for every other test (iodine stays brown, Benedict's stays blue, biuret stays blue).
9. **Benedict's tests for reducing sugars, not starch** — the colour reached (green, yellow or orange-red) shows how much reducing sugar is present.
10. **Use the command words** — identify (name, select or recognise), give (produce an answer from a given source or from recall), describe (state the points of a topic or give its main features). For "describe how you could test", give the reagent, what you do and the result.

Textbook Questions with Answers

Question B3.01

Name *one* carbohydrate that is found in both plants and animals.

Answer: Glucose. Plants make glucose in photosynthesis, and both plants and animals use glucose in respiration to release energy; in humans it is the form in which carbohydrate is transported in the blood plasma. (Starch is found only in plants and glycogen only in animals, so neither of these is a correct answer.)

Question B3.02

Animals that live in very cold places often have especially thick layers of fat underneath the skin (Figure B3.08). Suggest why.

(Figure B3.08: Bearded seals have a really thick layer of fat underneath their skin.)

Answer: The thick layer of fat under the skin acts as a heat insulating layer. Fat is a poor conductor of heat, so it reduces the amount of heat lost from the animal's warm body to its very cold surroundings, helping it to keep its body temperature high and constant. The fat is also an energy store: when food is scarce, the animal can use it in respiration to release energy, including heat to keep warm.

Question B3.03

Fat stores contain more energy per gram than carbohydrate stores. Suggest why birds that are about to go on long migrations build up stores of fat before they fly.

Answer: Flying a long distance needs a lot of energy, which is released by respiration in the bird's flight muscles, and the bird may not be able to feed for long periods during the journey, so it must carry its fuel with it. Fat stores contain more energy per gram than carbohydrate stores (roughly twice as much), so storing fuel as fat gives the bird the most energy for the least extra body mass. A lighter fuel store means the bird uses less energy carrying it, so it can fly further before it needs to stop and feed.

Question B3.04

Which element is found in all proteins, but not in carbohydrates and fats?

Answer: Nitrogen (N). All proteins are made of amino acids, which contain carbon, hydrogen, oxygen and nitrogen, but carbohydrates and fats contain only carbon, hydrogen and oxygen. (Sulfur is found in some proteins, but not in all of them, so it is not the answer.)

Practice Exam Questions with Answers

Practice Question 1

Identify which elements are contained in starch. (1 mark)

A) carbon and hydrogen only

B) carbon, hydrogen and oxygen

C) carbon, hydrogen and nitrogen

D) carbon, oxygen and nitrogen

Answer: B — carbon, hydrogen and oxygen. [1] Starch is a carbohydrate made of many glucose molecules ($C_6H_{12}O_6$), and carbohydrates contain only C, H and O. A is wrong because it leaves out oxygen; C and D are wrong because starch contains no nitrogen (nitrogen is found in proteins).

Practice Question 2

The table shows the results of tests on four samples of food. Identify which food contains *only* fat. (1 mark)

Food	Iodine test	Benedict's test	Ethanol emulsion test	Biuret test
A	brown	blue	milky white	blue
B	blue-black	red	colourless	purple
C	brown	blue	colourless	blue
D	blue-black	red	milky white	purple

A) Food A

B) Food B

C) Food C

D) Food D

Answer: A — food A. [1] Food A gives a positive ethanol emulsion test (milky white), so it contains fat, and every other test is negative: iodine stays brown (no starch), Benedict's stays blue (no reducing sugar) and biuret stays blue (no protein). Food B contains starch, reducing sugar and protein but no fat; food C contains none of these substances; food D contains starch, reducing sugar, fat and protein.

Practice Question 3

Glucose, starch and glycogen are carbohydrates. (7 marks)

a) Identify which substance(s) are found in plants only, and not in animals? [1]

b) Give the elements that the molecules of all these substances contain. [1]

c) Describe the structure of a starch molecule. [2]

d) Glucose is a reducing sugar. Describe how you could test a liquid to find out if it contains glucose. [3]

Answer:

a) Starch (only). [1] Glucose is found in both plants and animals, and glycogen is found only in animals.

b) Carbon, hydrogen and oxygen (C, H and O). [1]

c) A starch molecule is made of many glucose molecules [1] linked together in a long chain, which coils into a spiral shape [1].

d) Add Benedict's solution to a sample of the liquid in a test-tube [1]. Heat the tube in a water-bath (at about 80 °C) for a few minutes [1]. If glucose is present, the colour changes from blue to green, yellow or orange-red (brick red); if the mixture stays blue, there is no glucose [1].

Practice Question 4

This question is about the biuret test and the ethanol emulsion test. (6 marks)

a) Biuret reagent is used to test for proteins. These statements about the biuret test are incorrect. Identify the errors so that you can write corrected versions of each statement.

i) To carry out the biuret test, add biuret solution to the food and heat gently. [1]

ii) If a food contains protein, biuret reagent turns from purple to blue. [1]

b) Ethanol is used to test for fats and oils.

i) Describe how you would carry out the ethanol test. [3]

ii) Describe what you would see if the test gave a positive result. [1]

Answer:

a) i) The error is heating: the biuret test does not need heating. Corrected statement: to carry out the biuret test, add biuret solution to the food and shake to mix, without heating. [1]

a) ii) The colours are the wrong way round. Corrected statement: if a food contains protein, biuret reagent turns from blue to purple (violet). [1]

b) i) Cut the food into very small pieces and put them into a clean test-tube with a little ethanol [1]. Shake the tube so that any fats or oils dissolve in the ethanol [1]. Pour the ethanol, without any of the food, into another clean test-tube containing distilled water [1].

b) ii) A milky white (cloudy white) emulsion forms in the water. [1]

Summary of Key Concepts

From the Chapter Summary

- Carbohydrates are made of the elements C, H and O. Sugars, such as glucose, have relatively small molecules. Starch and glycogen are made of many glucose molecules linked in a long chain.
- Starch can be detected using the iodine test. Benedict's solution is used to test for reducing sugars.
- Fats and oils are also made of the elements C, H and O. Another name for fats and oils is lipids. A lipid molecule is made of three fatty acid molecules combined with one glycerol molecule.
- The ethanol emulsion test is used to detect the presence of fats and oils.
- Proteins contain the elements C, H, O and N. A protein molecule is made of many amino acid molecules combined in a long chain. There are 20 different amino acids, and their specific sequence in the molecule determines the specific protein that is formed.
- Biuret reagent is used to test for proteins.

Self-Evaluation Checklist

All of these skills belong to topic B3.01. Make sure you can:

- List the chemical elements that make up carbohydrates, fats and proteins
- State that starch and glycogen are made from glucose
- State that fats and oils are made from fatty acids and glycerol
- State that proteins are made from amino acids
- Describe how to use the Benedict's test for reducing sugars
- Describe how to use the iodine test for starch
- Describe how to use the biuret test for proteins
- Describe how to use the ethanol emulsion test for fats and oils

Exam Technique Tips

- For any "describe how you could test" question, include four things: the reagent, what you do (heat in a water-bath only for the Benedict's test), the colour for a positive result and the colour for a negative result
- Match the number of points to the marks: a 3-mark "describe" question needs three separate, clear points
- For an "identify" question, a single word or letter is enough; for a "give" question, a short answer from memory is enough
- With a results table, decide whether each test is positive or negative before you choose an answer
- Use precise colour words: blue-black, orange-red (brick red), violet (purple) and milky white; avoid vague phrases such as "changes colour" or "goes darker"
- Do not confuse glycogen (a carbohydrate energy store made from glucose) with glycerol (part of a fat molecule)